Hochschule für Technik
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Computer Networks

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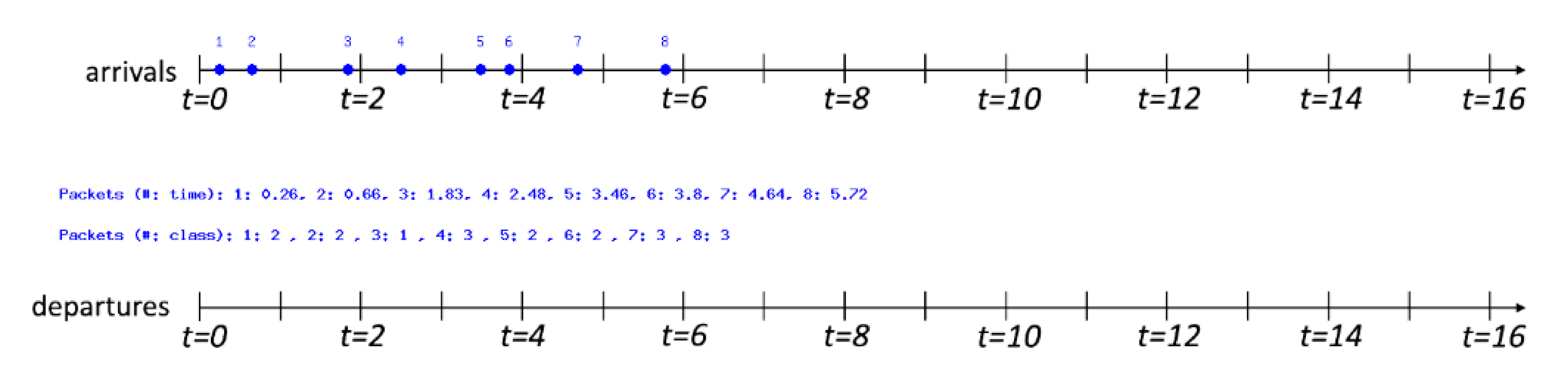
Lab-6

Packet Scheduling

Consider the arrival of 12 packets to an output link at a router in the interval of time [0, 6], as indicated by the figure below. We'll consider time to be "slotted", with a slot beginning at t = 0, 1, 2, 3, etc. Packets can arrive at any time during a slot, and multiple packets can arrive during a slot. At the beginning of each time slot, the packet scheduler will choose one packet, among those queued (if any), for transmission according to the packet scheduling discipline. Each packet requires exactly one slot time to transmit, and so a packet selected for transmission at time t, will complete its transmission at t+1, at which time another packet will be selected for transmission, among those queued. In the case of Priority and RR there will be three classes of traffic (1, 2, 3), with lower class numbers having higher priority in the case of priority schedule, or beginning earlier in the case of RR .
which packet will be sent at time 1, 2, 3, 4, 5, 6, 7, 8. According to the below scheduling discipline?

1. FIFO scheduler
2. priority
3. Round Robin



Packet Scheduling

Consider the arrival of 12 packets to an output link at a router in the interval of time [0, 6], as indicated by the figure below. We'll consider time to be "slotted", with a slot beginning at t = 0, 1, 2, 3, etc. Packets can arrive at any time during a slot, and multiple packets can arrive during a slot. At the beginning of each time slot, the packet scheduler will choose one packet, among those queued (if any), for transmission according to the packet scheduling discipline. Each packet requires exactly one slot time to transmit, and so a packet selected for transmission at time t, will complete its transmission at t+1, at which time another packet will be selected for transmission, among those queued. In the case of Priority, RR, and WFQ there will be three classes of traffic (1, 2, 3), with lower class numbers having higher priority in the case of priority schedule, or beginning earlier in the case of RR.
which packet will be sent at time 1, 2, 3, 4, 5, 6, 7, 8 according to the below scheduling discipline?

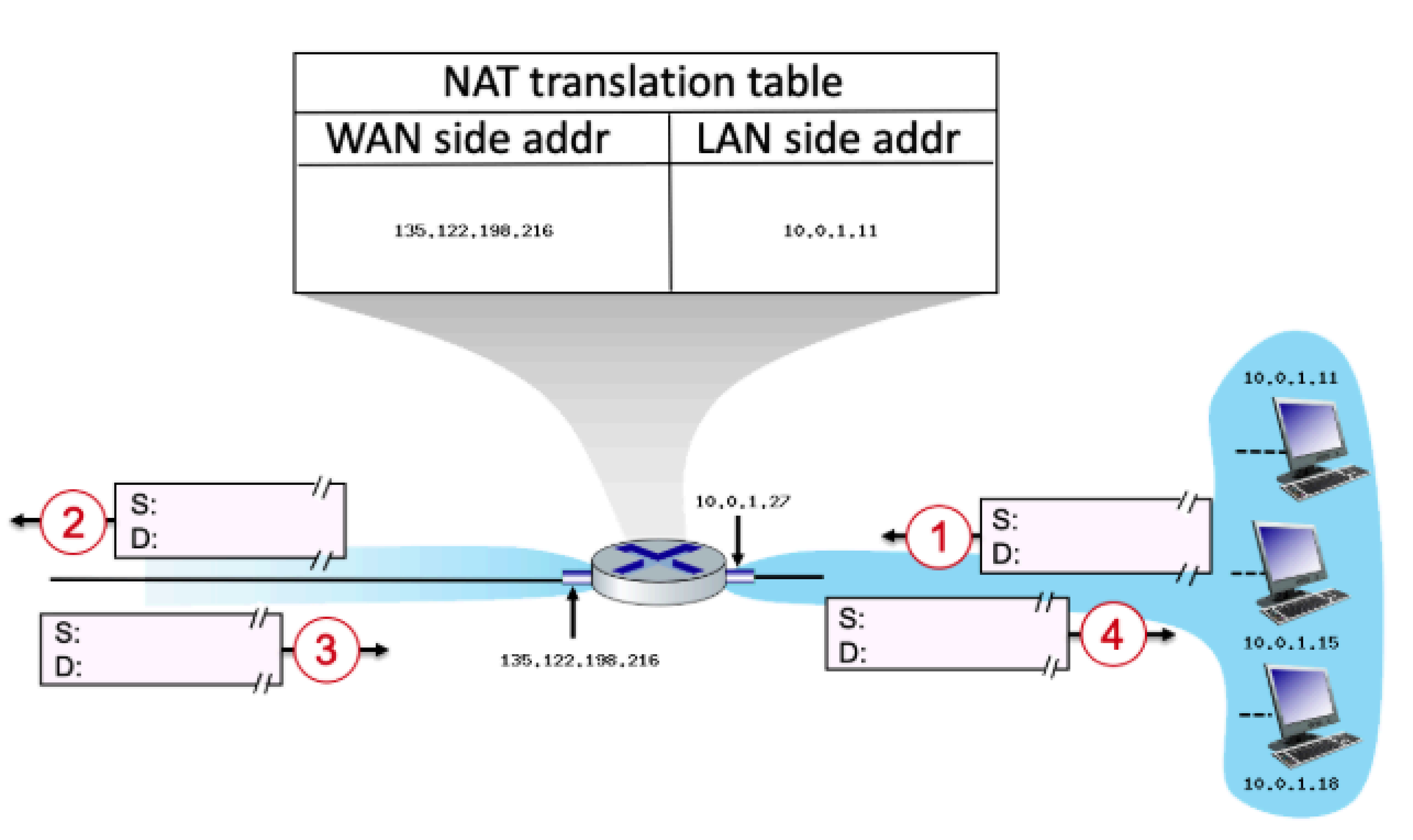
1. FIFO scheduler: 1, 2, 3, 4, 5, 6, 7, 8.
2. priority: 1, 3, 2, 5, 6, 4, 7, 8.
3. Round Robin: 1, 3, 2, 4, 5, 7, 6, 8.

Notiz: Auf der Companion-Website sieht die Lösung unter RR anderes aus. Ich vermute einen Fehler. Bitte lösen Sie die Aufgabe so wie wir es in der Übung gemacht haben und wie es hier in der Lösung in Punkt 3. ist.

Network Address Translation

Consider the scenario below in which three hosts, with private IP addresses 10.0.1.11, 10.0.1.15, 10.0.1.18 are in a local network behind a NAT'd router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router’s interface on the LAN side has IP address 10.0.1.27, while the router’s address on the Internet side has IP address 135.122.198.216. Suppose that the host with IP address 10.0.1.11 sends an IP datagram destined to host 128.119.171.185. The source port is 3325, and the destination port is 80.

1. Consider the datagram at step 1, after it has been sent by the host but before it has reached the router. What is the source IP address for this datagram?
2. At step 1, what is the destination IP address?
3. Now consider the datagram at step 2, after it has been transmitted by the router. What is the source IP address for this datagram?
4. At step 2, what is the destination IP address for this datagram?
5. Will the source port have changed? Yes or No.
6. Now consider the datagram at step 3, just before it is received by the router. What is the source IP address for this datagram?



7. At step 3, what is the destination IP address for this datagram?
8. Last, consider the datagram at step 4, after it has been transmitted by the router but before it has been received by the host. What is the source IP address for this datagram?
9. At step 4, what is the destination IP address for this datagram
10. Has a new entry been made in the router's NAT table? Yes or No.

SOLUTION

1. The source address will be the local host's IP, which is 10.0.1.11

2. The destination address will be the remote machine's IP, which is 128.119.171.185

3. The source address will be the router's public IP, which is 135.122.198.216

4. The destination address will be the remote machine's IP, which is 128.119.171.185

5. Yes, the NAT will change the source port.

6. The source address will be the remote machine's IP, which is 128.119.171.185

7. The destination address will be the router's public IP, which is 135.122.198.216

8. The source address will be the remote machine's IP, which is 128.119.171.185

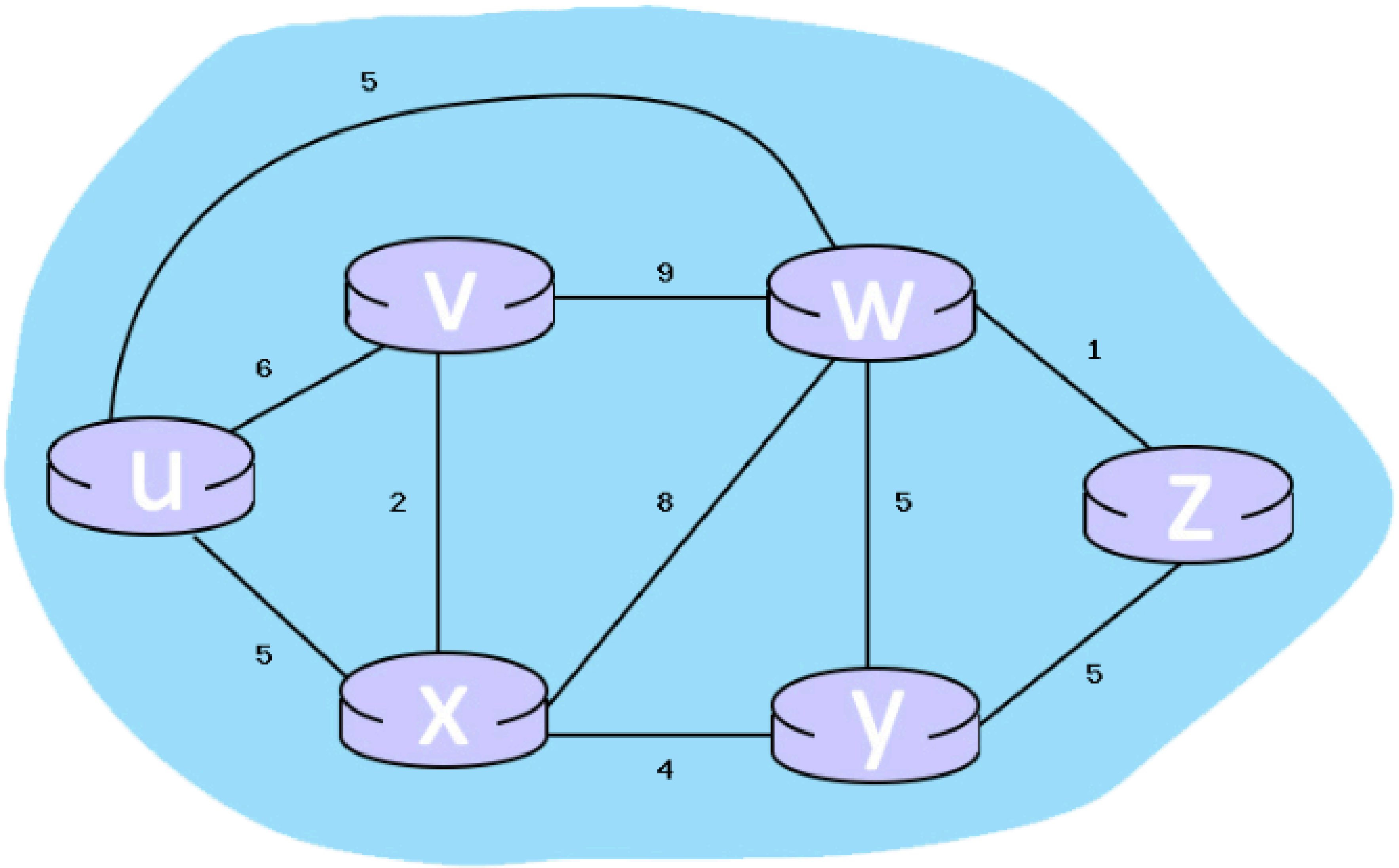
9. The destination address will be the local host's IP, which is 10.0.1.11

10. No, an entry is made when there's an outbound request, which only happens between step 1 and step 2.

Dijkstra's Link State Algorithm

Consider the 6-node network shown below, with the given link costs. Using Dijkstra's algorithm, find the least cost path from source node U to all other destinations and answer the following questions:

1. What is the shortest distance to node w and what node is its predecessor? Write your answer as n,p
2. What is the shortest distance to node z and what node is its predecessor? Write your answer as n,p
3. What is the shortest distance to node y and what node is its predecessor? Write your answer as n,p



SOLUTION

1. The minimum distance from node u to node w is 5, and node w's predecessor is node u. The full answer was: 5,u

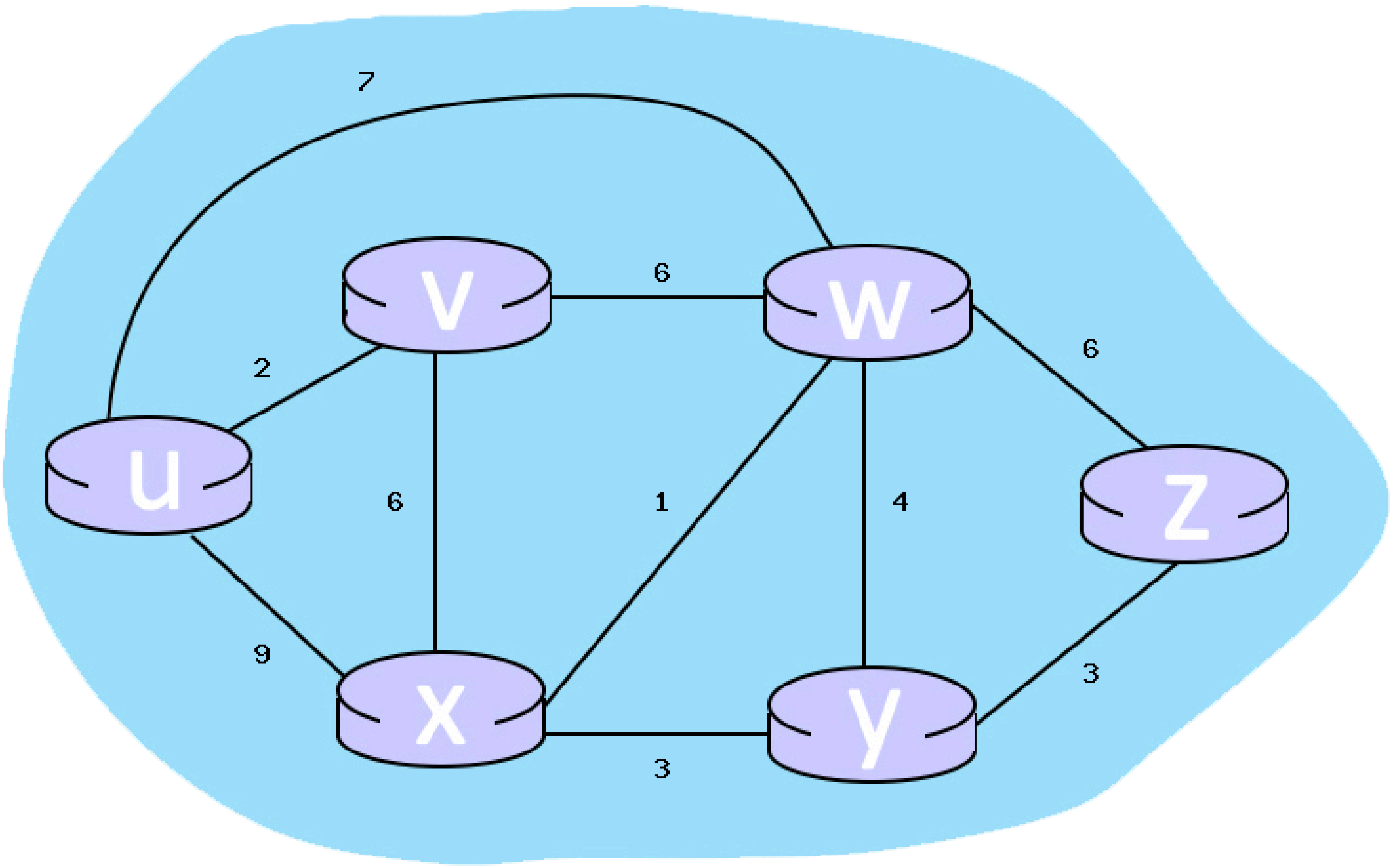
2. The minimum distance from node u to node z is 6, and node z's predecessor is node w. The full answer was: 6,w

3. The minimum distance from node u to node y is 9, and node y's predecessor is node x. The full answer was: 9,x

Dijkstra's Link State Algorithm 2

Consider the 6-node network shown below, with the given link costs. Using Dijkstra's algorithm, find the least cost path from source node U to all other destinations and answer the following questions:

1. What is the shortest distance to node z and what node is its predecessor? Write your answer as n,p
2. What is the shortest distance to node v and what node is its predecessor? Write your answer as n,p
3. What is the shortest distance to node x and what node is its predecessor? Write your answer as n,p



SOLUTION

1. The minimum distance from node u to node z is 13, and node z's predecessor is node w. The full answer was: 13,w

2. The minimum distance from node u to node v is 2, and node v's predecessor is node u. The full answer was: 2,u

3. The minimum distance from node u to node x is 8, and node x's predecessor is node v. The full answer was: 8,v